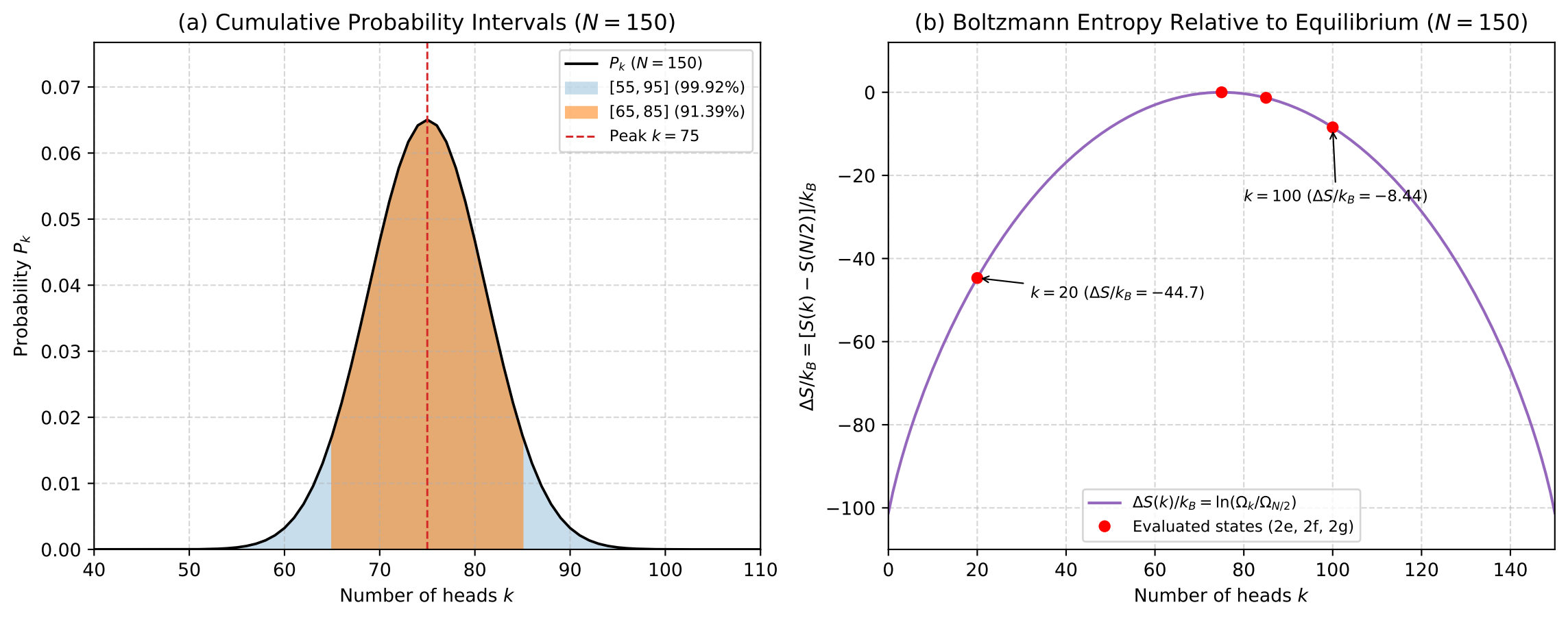
(a) Cumulative Probability Intervals ($N = 150$)
P_k ($N = 150$)
[55, 95] (99.92%)
[65, 85] (91.39%)
Peak $k = 75$
Probability P_k
Number of heads k
(b) Boltzmann Entropy Relative to Equilibrium ($N = 150$)
$\Delta S/k_B = [S(k) - S(N/2)]/k_B$
$k = 100$ ($\Delta S/k_B = -8.44$)
$k = 20$ ($\Delta S/k_B = -44.7$)
$\Delta S(k)/k_B = \ln(\Omega_k/\Omega_{N/2})$
Evaluated states (2e, 2f, 2g)
Number of heads k